# Allan Vikiru

## Machine Learning | Edge Computing | Conservation Technology

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GitHub | LinkedIn | Google Scholar

## Career Profile

- A junior researcher of Computer Science interested in PhD opportunities that explore the intersection of internet-of-things and machine learning to develop remote sensing or edge analytics applications.
- Knowledgeable in underlying technologies of edge AI, with Python (scikit-learn), R, Arduino and ESP32 as key technologies and tools.
- Demonstrates progressive research experience with engagement in 4 research publications and 3 projects that involved development of IoT architectures and machine learning models.

## Skills and Competencies

**Geospatial Analysis and Machine Learning**
Geostatistical Modelling (Kriging, Hybrid Kriging), Classical ML (Regression, Classification, Clustering, Association Analysis), Neural Networks

**Software Engineering**
Django, Git, Node.js, MATLAB, Python (NumPy, pandas, scikit-learn, rasterio, geopandas, GDAL), QGIS, R, Streamlit

**Professional Skills**
Technical Writing, Academic Presentations, Project Management, Stakeholder Engagement

## Education

**MSc. Information Technology (Business Intelligence & Data Analytics Option) [graduated June 2024]**
Strathmore University - Nairobi, Kenya
GPA – 3.64/4 (A)

**BSc. Informatics & Computer Science [graduated June 2021]**
Strathmore University - Nairobi, Kenya
GPA – 3.5/4 (A)

## Publications

**Conference and Journal Papers**

1. **Vikiru, A.,** Khakata, E.: A Wearable Sensor System for Monitoring the Risk of Fireground Injuries Among Firefighters. In: Yang, X.-S., Sherratt, R.S., Dey, N., and Joshi, A. (eds.) Proceedings of 11th International Congress on Information and Communication Technology. pp. 117–131. Springer Nature Switzerland, Cham (2027). https://doi.org/10.1007/978-3-032-36036-6_12.
2. Simiyu, D., Muchiri, H., **Vikiru, A**., Butime, J., Muti, Z.: A Forest Acoustics – Temporal Frequency Convolution Neural Network Model for Detecting Illegal Logging Activities in Forest. In: 2024 5th International Conference on Smart Sensors and Application (ICSSA). pp. 1–6 (2024). https://doi.org/10.1109/ICSSA62312.2024.10788639.
3. Ayankoso, S., Wang, Z., Shi, D., Yang, W., **Vikiru, A.,** Kamau, S., Muchiri, H., Gu, F.: Development of Long-Range, Low-Powered and Smart IoT Device for Detecting Illegal Logging in Forests. Journal of Dynamics, Monitoring and Diagnostics. (2024). https://doi.org/10.37965/jdmd.2024.550.
4. Simiyu, D., **Vikiru, A.,** Muchiri, H., Gu, F., Butime, J.: A Chainsaw-Sound Recognition Model for Detecting Illegal

Logging Activities in Forests. In: Ball, A.D., Ouyang, H., Sinha, J.K., and Wang, Z. (eds.) Proceedings of the UNIfied Conference of DAMAS, IncoME and TEPEN Conferences (UNIfied 2023). pp. 797–806. Springer Nature Switzerland, Cham (2024). https://doi.org/10.1007/978-3-031-49421-5_65.

**Datasets and Preprints**

5. **Vikiru, A**., Simiyu, D., Kahoro, S., Muti, Z., Kamau, S., Macharia, A., Zemp, D.C., Muchiri, H.: Kenya Forest Sounds Dataset. [Dataset], (2026). https://doi.org/10.5281/zenodo.21443074.
6. **Vikiru, A.,** Muiruri, M., Ateya, I.: An Overview on Cloud Distributed Databases for Business Environments [Preprint] (2023). https://doi.org/10.48550/arXiv.2301.10673.

**Theses**

7. **Vikiru, A**.: A Predictive Model for Determining Vegetable-derived Macronutrients for a Diabetic Patient (2024). https://hdl.handle.net/11071/16545.

## Research Experience

**Curbing Illegal Logging Practices using Sound Based Detection Techniques [July 2023 – December 2024]**

- Supported by Climate Change AI Grants 2023, this project involved creating an edge AI architecture in forested environments to determine logging sounds by chainsaw and axe as a detective measure for illegal logging activities. This research project was conducted under supervision of Dr Henry Muchiri (Strathmore University).
- My technical roles in this project were focused on developing the base classifier for acoustic classification (*accuracy of 87%*) and implementation and real-world testing of the edge node device and complementary LoRaWAN architecture.
- Led in the collection, analysis and publication of forest sound data (*5500 clips published in July 2026*) and handled administrative tasks by engaging project partners through coordinating meetings and field visits; and writing grant proposals and documentations.

**A Predictive Model for Determining Vegetable Derived Macronutrients for a Diabetic Patient [May 2023 – June 2024]**

- Supported by the National Research Fund (Kenya), this work featured the development of a regression-based geospatial map and recommendation-based tool that can be used to determine suitable plant vegetables for diabetic patients in Kenya based on the distribution of soil macronutrients. The project served as the thesis topic in fulfillment of the MSc. Information Technology degree under supervision of Prof. Vincent Omwenga (Strathmore University).
- In this, geospatial maps (*lowest RMSE – 0.37, 2.34*) were developed through geostatistical, machine learning and hybrid (geostatistical and machine learning techniques) and applied to a recommendation-based tool that could be used to determine suitable plant vegetables for diabetic patients in Kenya based on the distribution of soil macronutrients.
- My roles in this project were the geospatial modelling of georeferenced soil maps; development of an ensemble clustering-classification recommendation model; proposal writing and project documentation.

**Detecting the Risk of Fireground Injuries among Firefighters using IoT [April 2020 – September 2021]**

- This study looked at developing a wearable device that firefighters can wear in the line of duty, to sense their vitals such as heart rate and blood oxygen and environmental parameters such as atmospheric temperature to determine the risk of sustaining fireground injuries. It was the final year project of the BSc. Informatics & Computer Science degree under supervision of Dr. Esther Khakata (Strathmore University).
- My roles in this project were in IoT prototyping: design and implementation of the node wearable device and complementary GSM architecture. Regression analysis of node-collected data (*R-squared of 0.44*) was also conducted together with the implementation of a web application to present real-time results.
- In the early stages of dissemination, this project was presented in a national IoT and AI prototyping contest and received recognition as a Top 10 finalist project. It was then restructured as a research paper for presentation and publication at an international conference.

## Work Experience

**Teaching Fellow – School of Computing and Engineering Sciences, Strathmore University [July 2024 – Present]**

- Teaching and assessing undergraduate students in these courses so far: Artificial Intelligence Concepts, Embedded Systems & IoT, Software Engineering & Information Security, Data Governance & Cloud Computing.
- Continually supervising undergraduate students on their final-year projects on systems involving deep learning (NLP, Computer Vision applications) and IoT as key technologies. Also serving as a project supervisor and practitioner of Challenge-Driven Education as a methodology for teaching and learning.

**Edge AI Researcher & Co-lead – Conservation Technology Research Hub, Strathmore University [September 2025 – Present]**

- Coordinating a team of 6 faculty researchers and at least 10 undergraduate and graduate students on rolling bases in projects on bioacoustics analysis (neural network modelling), edge node prototyping, and UAV/drone implementation.
- Conducting administrative activities on grant proposal writing, project publication and dissemination, external partner engagement, and developing training programmes.

**Graduate Assistant – School of Computing & Engineering Sciences, Strathmore University [March 2022 – June 2024]**

- Supported senior faculty in teaching and research activities in facilitating laboratory sessions, assessing coursework and developing research proposals.

**Software Developer Intern – IoT & Wireless Networks Lab, @iLabAfrica [January 2022 - March 2022]**

- Designed and implemented breadboard models for node devices, and a monitoring dashboard in Vue.js and Node.js with a Keycloak authentication system for a research project that involved LoRaWAN communication.

## Certifications & Licenses

**Remote Pilot License**
Kenya Civil Aviation Authority
Issued January 2025

## Conferences & Workshops

**Delegate – GDH Partner Days on Challenge Driven Education [March 2026]**
Royal Institute of Technology – Stockholm, Sweden

**Presenter - A Wearable Sensor System for Monitoring the Risk of Fireground Injuries among Firefighters [February 2026]**
11th International Congress on Information and Communication Technology - London, United Kingdom

**Delegate – Microsoft Faculty Skilling and Immersion Program (Cohort 3 – AI-Oriented Education) [March - June 2025]**
Microsoft ADC & Microsoft Leap - Nairobi, Kenya

**Delegate - 5th International Conference on Smart Sensors and Application [September 2024]**
Universiti Teknologi Malaysia - Penang, Malaysia

**Delegate – CMU Innovation Summit 2024 [January 2024]**
Carnegie Mellon University Africa - Kigali, Rwanda

**Organiser & Presenter – Fundamentals of Raspberry Pi 400 [November 2023]**
Strathmore University - Nairobi, Kenya